Health Insurance Policy Term Project

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Abstract

This study analyzes U.S. county-level health insurance data from 2017 to 2022 to uncover the impact of socio-demographic factors on enrollment trends. Using comprehensive data and advanced statistical tools such as SAS, the report examines predictors including age, disability status, household income (adjusted for inflation and regional variations), educational attainment, and living arrangements. The findings reveal stark disparities influenced by socio-economic status, with coverage rates varying significantly based on education level, income brackets, and disability status. Results also show notable regional patterns, underscoring the role of local policies and economic conditions in shaping access to insurance. By offering targeted policy recommendations, this report contributes actionable insights aimed at fostering equitable health insurance coverage nationwide.

Introduction

Within the complex society that is today's world, health insurance policies play a pivotal role in ensuring access to essential healthcare services and providing financial protection against unforeseen medical expenses. Analyzing health insurance data has become increasingly critical for policymakers, insurers, and healthcare providers to identify trends, optimize coverage, and address gaps in accessibility and affordability. This report explores the intricate patterns and key metrics within U.S. county-level health insurance data from 2017 to 2022, focusing on enrollment demographics. It examines the effects of factors such as age, poverty level, educational attainment, household income, work experience, and disability status.

The analysis incorporates a range of socio-demographic categories, including age (under 19, 19 to 64, and 65 years or older), disability status (with or without disability), poverty level (138% to 399% poverty, at or above 400% poverty, below 138% Poverty, and below 100% Poverty), educational attainment is another critical factor, with categories ranging from less than a high school graduate to bachelor's degree or higher. Additionally, work experience includes those who did not work, worked part-time, and those who worked full-time. These categories provide a nuanced view of how diverse demographic factors influence health insurance participation.

Understanding these dynamics is essential for informing policy decisions and improving health insurance systems to better meet the needs of diverse populations. For example, identifying disparities in coverage among different age groups or education levels can guide targeted interventions and resource allocation. By controlling for regional variations and inflation-adjusted income, the analysis seeks to offer an equitable and precise understanding of enrollment patterns. This approach ensures that findings are not only statistically robust but also relevant for addressing real-world challenges in the healthcare domain.

Ultimately, this report aims to contribute to the broader dialogue on health equity and policy reform. By examining the interplay between socio-demographic factors and health insurance enrollment, it provides actionable insights to help stakeholders design policies that promote accessibility, affordability, and inclusivity in health coverage for all Americans. The findings will be instrumental in highlighting areas for improvement and creating a foundation for future research into health insurance dynamics.

Theoretical Framework

The theoretical foundation of this analysis rests on several interrelated concepts that explain the socio-economic and structural factors influencing health insurance enrollment. At its core is the concept of the socioeconomic gradient in health, which posits that individuals with higher socioeconomic status (SES) experience better health outcomes and greater access to healthcare resources (Keisler, 2022). SES encompasses both income and educational attainment, which directly influence an individual's ability to afford health insurance premiums, access employer-sponsored plans, and navigate complex insurance markets. This framework highlights the disparities faced by lower SES groups and underscores the importance of targeted interventions.

Another key theory informing this study is behavioral economics, which explores how individuals make decisions about health insurance based on perceived costs, benefits, and constraints. For many individuals, especially those in lower income brackets, the financial burden of insurance premiums and deductibles outweighs the perceived benefits, leading to the decision to remain uninsured (Carmen, 2015). Behavioral economics also accounts for the role of cognitive biases, such as a lack of information or understanding about the long-term benefits of coverage, in shaping enrollment trends.

The concept of intersectionality is central to understanding how overlapping identities and disadvantages influence access to health insurance (Vardell, 2019). For example, a single parent with limited education and a disability faces compounded barriers that make obtaining coverage far more challenging than for someone with only one of these characteristics. Intersectionality emphasizes that policies aimed at increasing health insurance enrollment must account for the multifaceted experiences of individuals, rather than treating each demographic variable in isolation.

Institutional theories, particularly those related to the expansion of public insurance programs, provide another lens through which to examine enrollment trends. States that expanded Medicaid under the Affordable Care Act (ACA) have seen significant increases in coverage among low-income populations (Cha, 2022). This highlights the role of institutional support in mitigating socio-economic barriers and suggests that policy frameworks must adapt to local economic and political conditions to maximize their effectiveness.

Finally, the dynamics of risk pooling in insurance markets explain how socio-demographic factors influence the overall stability of health insurance systems. Younger, healthier individuals are less likely to enroll in insurance, which can destabilize risk pools and drive up costs for high-risk populations (Buttgens, 2023). This framework emphasizes the importance of policies that encourage broad participation to maintain affordable premiums and ensure the sustainability of the health insurance market.

Together, these theories provide a comprehensive framework for analyzing the complex interplay of socio-demographic variables and systemic structures in determining health insurance enrollment. They underscore the need for multifaceted, data-driven approaches to address disparities and promote equitable access to healthcare.

Methodology and Data

This analysis relies on a robust dataset comprising U.S. county-level health insurance data from 2017 to 2022. The dataset includes detailed information on enrollment rates and socio-demographic characteristics, offering a granular view of health insurance dynamics across diverse geographic and economic contexts. Additional data sources, such as Census Bureau reports and state Medicaid expansion statistics, were used to contextualize the findings and validate key trends.

The variables examined in this study include age, disability status, household income (adjusted for inflation), educational attainment, work experience, and poverty levels. Age groups were categorized into under 19, 19 to 64, and 65 years or older to account for key life stages and eligibility for public insurance programs like Medicare. Disability status was analyzed as a binary variable, while household income was adjusted for regional cost-of-living differences and divided into quintiles. Educational attainment was categorized into four levels, ranging from less than high school to a bachelor's degree or higher. Work experience was arranged through those who did not work, worked part-time, and worked full-time. Lastly, poverty levels were presented as a whole with those at 138% to 399% poverty, at or above 400% poverty, below 138% poverty, and below 100% poverty, to get a more wholesome picture of how varying poverty levels are impacted by health insurance coverage.

Methodologically, the study employed a combination of descriptive and inferential techniques. Descriptive statistics provided a snapshot of enrollment trends across socio-demographic categories, highlighting disparities and regional variations. Detailed figures were used to provide a comprehensive view of each of our variables and their subcategories for insured and uninsured to effectively cross-compare. The analysis was conducted using SAS, leveraging these tools' capabilities for data cleaning, visualization, and statistical modeling. Geospatial analysis was also incorporated within our appendices, to map enrollment trends and identify regional patterns, providing a visual representation of disparities across the U.S. as additional support to our final analysis takeaways.

Presentation of Results

Table 1: Descriptive Statistics for Insured Individuals

Descriptive Statistics For Insured

The MEANS Procedure

Variable	N	Mean	Std Dev	Lower 95% CL for Mean	Upper 95% CL for Mean
InsuredWithDisability	19324	12108.25	32799.88	11645.76	12570.74
InsuredNoDisability	19324	79419.96	259104.50	75766.52	83073.40
InsuredLessThanHighSchool	19324	6124.80	26398.51	5752.57	6497.02
InsuredHighSchoolGraduate	19324	15598.63	41102.14	15019.08	16178.18
InsuredSomeCollege	19324	17556.01	52382.36	16817.40	18294.61
InsuredBachelorOrHigher	19324	21118.99	76532.93	20039.86	22198.12
InsuredUnder19	19324	23111.29	74546.80	22060.16	24162.42
Insured19To64	19324	52815.08	173560.69	50367.83	55262.33
Insured65AndOver	19324	15601.84	44586.21	14973.17	16230.52
InsuredWorkedFullTime	19323	29676.39	97192.95	28305.91	31046.87
InsuredWorkedLessThanFullTime	19323	12753.16	42730.01	12150.64	13355.68
InsuredDidNotWork	19323	10387.30	34602.63	9899.38	10875.22
InsuredBelow138Poverty	19323	16859.52	56666.52	16060.49	17658.55
Insured138To399Poverty	19323	35769.15	109894.38	34219.57	37318.73
InsuredAbove400Poverty	19323	37805.29	128149.69	35998.31	39612.28
InsuredBelow100Poverty	19323	11389.08	38410.33	10847.47	11930.69
InsuredIncomeUnder25K	19323	11760.62	36291.92	11248.88	12272.36
InsuredIncome25KTo50K	19323	15680.35	47443.54	15011.36	16349.33
InsuredIncome50KTo75K	19323	14945.53	44695.16	14315.30	15575.76
InsuredIncome75KTo100K	19323	12539.47	37971.41	12004.05	13074.89
InsuredIncomeOver100K	19323	35494.97	129385.10	33670.56	37319.38

Table 2: Descriptive Statistics for Uninsured Individuals

Descriptive Statistics For Uninsured

The MEANS Procedure

Variable	N	Mean	Std Dev	Lower 95% CL for Mean	Upper 95% CL for Mean
UninsuredWithDisability	19324	739.5253053	2227.99	708.1100948	770.9405158
UninsuredNoDisability	19324	8437.40	34818.61	7946.45	8928.35
UninsuredLessThanHighSchool	19324	1712.31	9114.09	1583.80	1840.82
UninsuredHighSchoolGraduate	19324	2214.52	7887.47	2103.30	2325.73
UninsuredSomeCollege	19324	1661.25	5963.26	1577.16	1745.33
UninsuredBachelorOrHigher	19324	865.8300559	3851.54	811.5223445	920.1377673
UninsuredUnder19	19324	1289.32	5275.72	1214.93	1363.71
Uninsured19To64	19324	7759.90	31302.27	7318.53	8201.27
Uninsured65AndOver	19324	127.7023908	699.0650825	117.8453991	137.5593825
UninsuredWorkedFullTime	19323	3177.91	13536.30	2987.04	3368.78
UninsuredWorkedLessThanFullTime	19323	2611.71	9760.11	2474.08	2749.33
UninsuredDidNotWork	19323	1970.51	8134.04	1855.82	2085.21
UninsuredBelow138Poverty	19323	3222.91	13154.80	3037.42	3408.40
Uninsured138To399Poverty	19323	4472.84	18451.19	4212.66	4733.01
UninsuredAbove400Poverty	19323	1413.38	5535.63	1335.33	1491.44
UninsuredBelow100Poverty	19323	2223.93	8874.29	2098.80	2349.07
UninsuredIncomeUnder25K	19323	1898.00	7342.81	1794.46	2001.54
UninsuredIncome25KTo50K	19323	2497.84	10274.45	2352.96	2642.71
UninsuredIncome50KTo75K	19323	1845.21	7575.29	1738.39	1952.03
UninsuredIncome75KTo100K	19323	1119.28	4720.45	1052.72	1185.85
UninsuredIncomeOver100K	19323	1722.33	7590.40	1615.30	1829.36

In beginning our analysis of US health insurance coverage between 2017 and 2022, we examined crucial socio-demographic variables among all 50 States with a collective descriptive statistics table displayed in Tables 1 and 2. This allowed us to understand the patterns within the categories of insured and uninsured individuals across the country. Using the county-level data from the US Census Bureau we utilized the variables of age, disability status, educational attainment, work experience, and poverty levels broken down into the predicted subcategories displayed in the tables. These categories are calculated based on the subcategories for both insured and uninsured consisting of N observations, average mean, standard deviation, lower, and upper 95% confidence intervals for mean across the years 2017 and 2022. This provided a nuanced perspective on the diverse factors that influence health insurance participation.

Moreover, in regard to key findings from the descriptive statistics analysis, we highlight a strong correlation between educational attainment and health insurance coverage, along with income levels on uninsured rates. Individuals that were placed into the subcategory among educational attainment with a bachelor's degree or higher had the highest mean for insurance coverage, which may reflect the increase in accessibility of sponsored health plans by employers with a higher earning job. However, those with less than a high school diploma are far more likely to be uninsured. Additionally, income level has an impactful relationship with those below the 138% poverty line, who experience higher uninsured rates, despite Medicaid programs that are supposed to target demographics as such. Lastly, age groups are revealed to have significant contrasts with children under 19 and seniors over the age of 65 having the highest insurance coverage due to programs like Medicare, whilst the working-age adults from ages 19 to 64 face the highest coverage gaps. Overall, these findings underscore the importance of addressing the various categorical and sub-categorical barriers most notably, low-income, working-age, and less-educated populations, that will continue to delve into throughout the rest of our paper.

Figure 1: Total Number of Insured Population in the US by Age Groups (2017-2022)

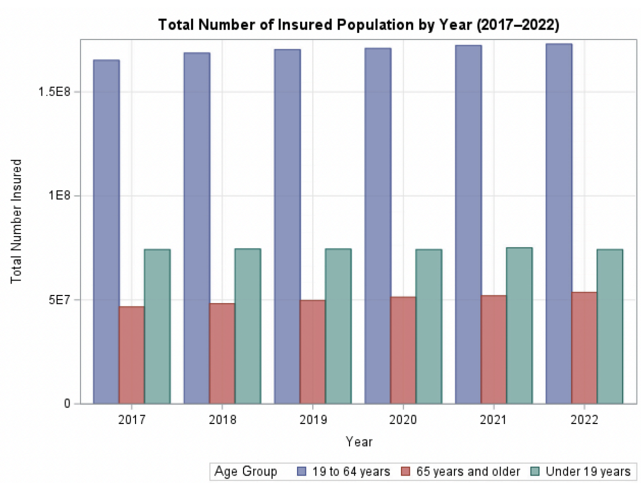
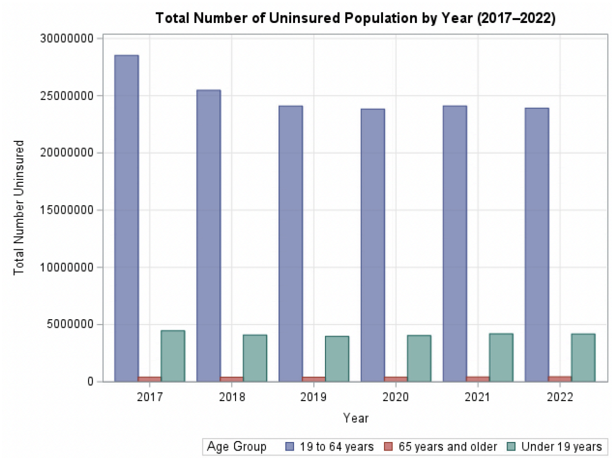


Figure 2: Total Number of Uninsured Population in the US by Age Groups (2017-2022)



Age Groups:

Figures 1 and 2 refer to the total number of insured and uninsured persons categorized by age groups: under 19, 19 to 64, and 65 and older. In Figure 1, the population aged 65 and older consistently exhibits the highest rates of being insured. This result could be attributed to the eligibility and widespread availability of Medicare for those aged 65 and up. The individuals aged 19-64, show a slight upward trend in insurance rates; This could be a reflection of the positive influence that ACA marketplace plans and Medicaid expansion have had on this group of individuals. Individuals 19 and under remained stable over the years, which could be explained by the Medicaid expansion and the CHIP coverage for children.

Alternatively, Figure 2 reflects the total number of uninsured individuals among the same age groups. From the chart, we can see that the 19 to 64 age group has the highest uninsured rates, which can be attributed to gaps in employer-sponsored insurance plans. Individuals aged 19 and under have low uninsured rates, but indicate a minor fluctuation from 2017-2022, which can reflect some gaps in pediatric Medicare coverage.

Figure 3: Total Number of Insured Population by Disability Status (2017-2022)

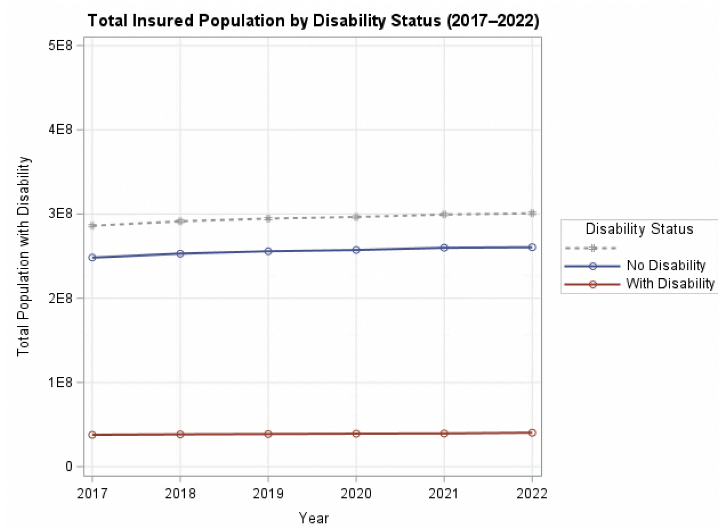


Figure 4: Total Number of Uninsured Population by Disability Status (2017-2022)

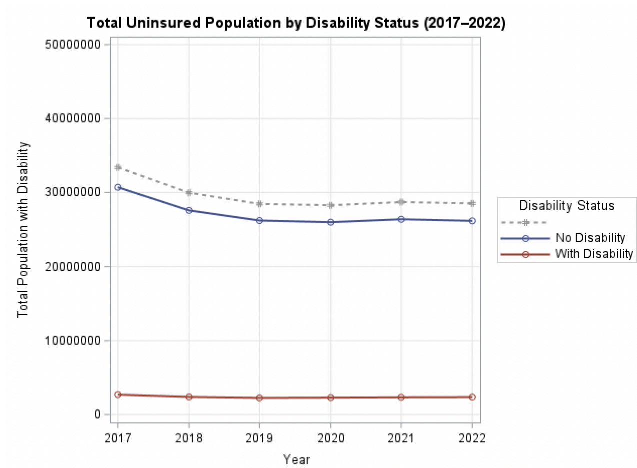
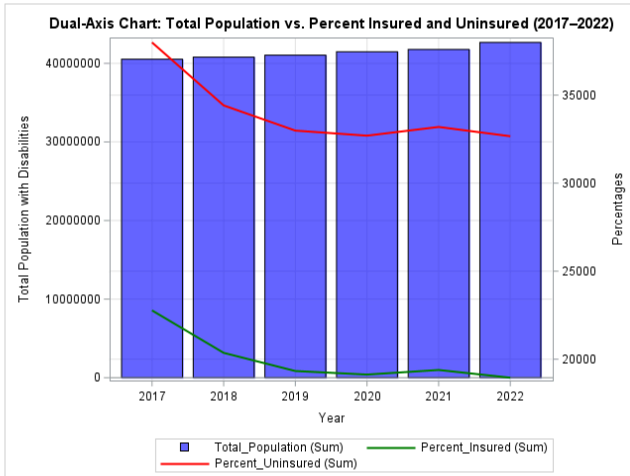


Figure 5: Dual-Axis Chart: Disability Total Population vs Percent Insured and Uninsured (2017-2022)



Disability Status:

Figures 3, 4, and 5 reflect varying visualizations of the total population who are insured and uninsured with or without a disability, as well as the total population. Figure 3, represents the total number of insured people based on their disability status. It can be seen that individuals with no disability have a gradually increasing trend of getting insured. In contrast, their counterparts in the graph i.e. people with disability have a constant number with a slight increment. While both groups show improvements in insured numbers, systemic challenges specific to the disabled population (e.g., fewer employment opportunities and reliance on government programs) may slow progress compared to their non-disabled counterparts.

Figure 4 represents the total number of uninsured people based on their disability status. It can be observed that people with no disability have a gradually decreasing trend of being uninsured. Similarly, individuals with disability also have a decreasing but very slight decreasing trend in not being uninsured. Figures 3 and 4 collectively highlight the difference in trend in healthcare coverage between individuals with and without disabilities, even though uninsured rates decline for both groups.

Figure 5 represents a dual-axis graph comparing 3 datasets: total population, percent insured, and present uninsured. This chart supports the information represented in Figures 3 and 4, and suggests an overall improvement in insurance coverage between 2017 and 2022, as reflected by decreasing uninsured rates and increasing insured rates. The reason for the increase in the total number of insured people can be due to policy reforms like the Affordable Care Act, Medicaid expansion, and government subsidies, which have made insurance more affordable and accessible over the years.

Figure 6: Total Number of Insured Population by Educational Attainment (2017-2022)

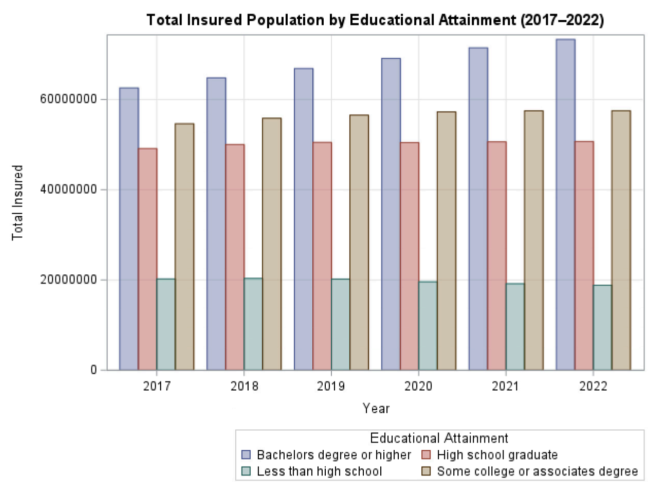
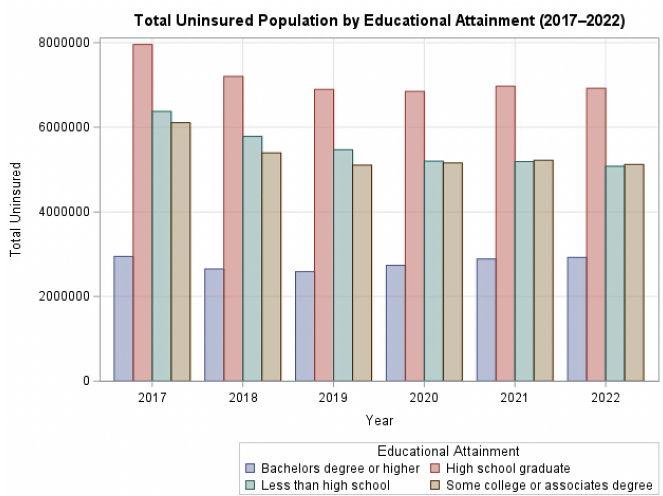


Figure 7: Total Number of Uninsured Population by Educational Attainment (2017-2022)



Educational Attainment:

Figure 6 illustrates the total number of insured individuals across different levels of educational attainment from 2017 to 2022. The data reveals a clear trend: individuals with higher education levels, particularly those holding a bachelor's degree or higher, consistently form the largest proportion of the insured population. Conversely, those with less than a high school education represent the smallest insured group. The insured population shows stability or slight growth across all educational groups over the years, suggesting that higher educational attainment is strongly associated with increased access to or likelihood of having insurance coverage.

Figure 7, on the other hand, focuses on the uninsured population segmented by educational attainment over the same period. The trends are nearly inverse to those seen in Figure 6. The largest share of the uninsured population consistently comes from individuals with less than a high school education, followed by high school graduates. Those with some college education or a bachelor's degree or higher represent a significantly smaller proportion of the uninsured. While the uninsured population shows minor fluctuations over time, the overarching pattern remains consistent, with individuals in lower educational attainment categories disproportionately lacking insurance.

Comparing the two figures, a clear disparity emerges between educational attainment and insurance coverage. Higher education levels strongly correlate with higher rates of insurance, while lower education levels correspond to higher uninsured rates. This trend reflects the role of education in securing stable, well-paying jobs that often include employer-sponsored insurance. Government programs like Medicare and Medicaid, along with the Affordable Care Act (ACA), have helped reduce uninsured rates, particularly for those with moderate education. However, individuals with less than a high school education still face significant barriers, including gaps in Medicaid eligibility, lack of awareness about ACA benefits, and employment in sectors without insurance coverage.

Figure 8: Dual-Axis Chart: Work Experience Trends For Insured (2017-2022)

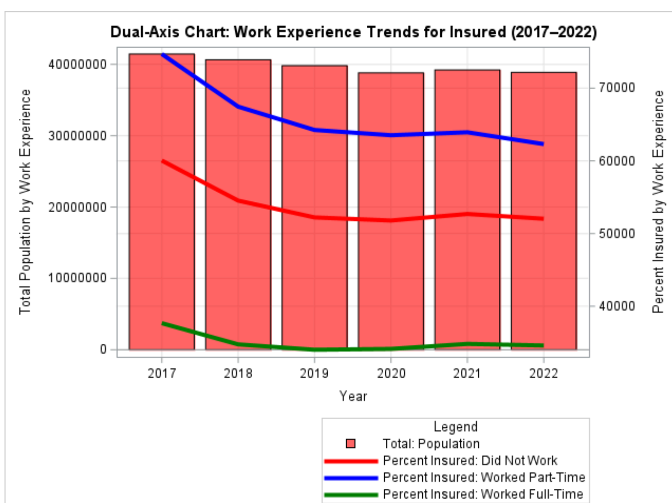
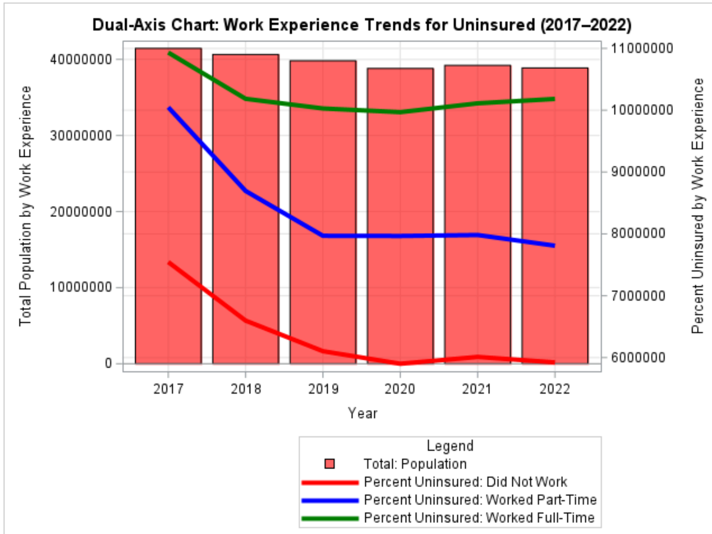


Figure 9: Dual-Axis Chart: Work Experience Trends for Insured (2017-2022)



Work Experience:

Figures 8 and 9, use a dual-axis chart to show the relationship between the total population, and the percentages of insured and uninsured groups based on their level of work experience from 2017 to 2022. The work experience levels are classified into three different categories: "Did Not Work" which refers to unemployed individuals, "Worked Part-Time" which refers to individuals who are employed but work less than 40 hours per week, and "Worked Full-Time" which refers to the individuals with full-time employment.

These charts highlight the trends amongst the population relative to their employment type. For insured individuals, the percentage of individuals who did not work showed a downward trend from 2017 to 2022. This might suggest that unemployed individuals are facing challenges in maintaining their health insurance coverage over time. The percentage of individuals who worked part-time, also showed a gradual decline, although less steep, this may indicate that individuals who work part-time may have limited access to employer-sponsored insurance, thus also having a difficult time maintaining their insurance coverage. For individuals who were employed full-time, their insurance rates remain consistently high, although showing a minor decline over the years after 2017. This result highlights the strong correlation between full-time employment and access to health insurance. This could be due to wider access to employer-sponsored health insurance plans, or a higher income level in the individuals. In summary, the declining trends for unemployed individuals and individuals who are employed part-time, compared to that of full-time individuals, suggests a growing gap in healthcare insurance access in America. Full-time workers maintain the highest insured rates from 2017-2022, which emphasizes the importance of full-time employment for health insurance access in America.

Amongst the uninsured, the percentage of uninsured individuals who were unemployed shows a downward trend in insurance rates. Although this seems a bit counter-intuitive, it can also explain efforts to provide public health coverage to unemployed individuals, like the expansion of Medicaid. Individuals who work Part-time, also see a downward trend, this result highlights the vulnerability of this group, due to a smaller access to affordable insurance options, as it makes up the largest portion of uninsured individuals. Individuals with full-time employment show relatively low rates of being uninsured. This once again highlights the importance of full-time employment for health insurance coverage. The declining rates amongst part-time and unemployed individuals can reflect a systematic effort to improve insurance through policy interventions like the Affordable Care Act and the Medicaid Expansion. However, Full-time workers still face proportionally lower uninsured rates, which further solidifies the importance of a full-time job for individuals in America.

Both of these figures highlight the importance of Full-time employment and its correlation with insurance rates. This result underscores the importance of employer-sponsored health insurance for Americans. Additionally, in both figures, Unemployed and Part-time individuals remain the most vulnerable groups to insurance coverage. This result highlights the widespread difficulties for affordable healthcare in America, especially for individuals with a lower income.

Figure 10: Comparison of Insured People by Income Level (2017-2022)

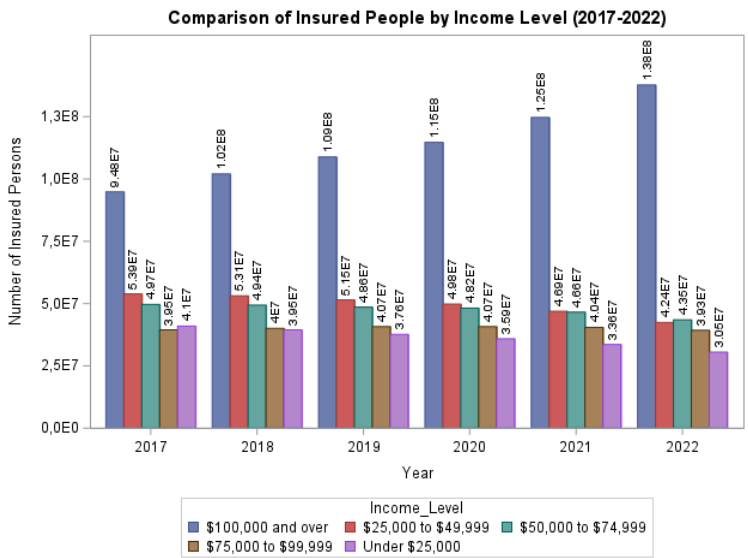
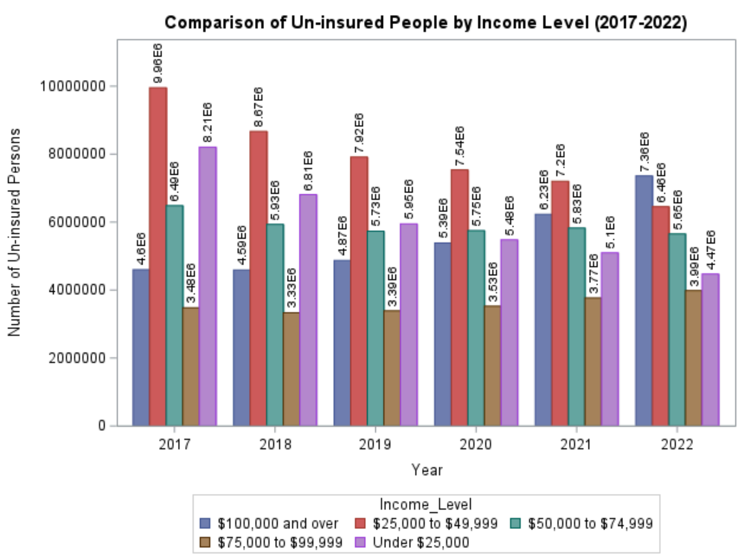


Figure 11: Comparison of Uninsured People by Income Level (2017-2022)



Income level:

Figures 10 and 11 provide a clear picture of health insurance coverage trends in the United States across different income levels from 2017 to 2022, showing significant patterns and disparities. Overall, the number of insured individuals has steadily increased over the years, while the number of uninsured individuals has gradually decreased. This reflects some progress in improving access to health insurance, although income-based inequalities remain a significant challenge. Among the income groups, individuals earning "$100,000 and over" show the highest levels of insurance coverage, increasing from about 94.8 million in 2017 to approximately 138 million in 2022. This group consistently exhibits strong access to health insurance, likely due to the availability of employer-sponsored plans and their financial ability to afford private insurance.

On the other hand, the number of uninsured people has been declining, but the reduction is not evenly distributed across income levels. High-income earners, particularly those in the "$100,000 and over" category, saw a significant drop in uninsured numbers, from 4.6 million in 2017 to 3.9 million in 2022. However, the lower-income groups, such as those earning "Under $25,000" and "$25,000 to $49,999," continue to face high levels of uninsured individuals. For instance, the "Under $25,000" group only showed a modest decline in uninsured numbers, from 9.9 million in 2017 to 9.4 million in 2022. This highlights the ongoing difficulties faced by economically disadvantaged populations in accessing affordable health coverage.

The middle-income groups, including "$50,000 to $74,999" and "$75,000 to $99,999," show moderate improvements in both insured and uninsured figures. Many in these groups benefit from employer-sponsored insurance; however, gaps remain for those who do not qualify for public programs like Medicaid or for subsidies under the Affordable Care Act. This suggests that while insurance accessibility has improved for these groups, there are still challenges for those caught between being ineligible for government assistance and unable to afford private insurance.

In conclusion, the overall trends reveal progress in expanding health insurance coverage, but they also highlight the persistence of disparities based on income. Higher-income groups enjoy almost universal coverage, whereas lower-income individuals remain disproportionately uninsured. These findings indicate the need for targeted policies to address these inequalities. Expanding Medicaid, increasing subsidies for low- and middle-income groups, and reducing the cost of private insurance could help close the coverage gap. The data clearly demonstrates the importance of addressing socioeconomic barriers to ensure equitable access to healthcare coverage for all income levels in the United States.

Figure 12: Total Number of Insured Population by Poverty Level (2017-2022)

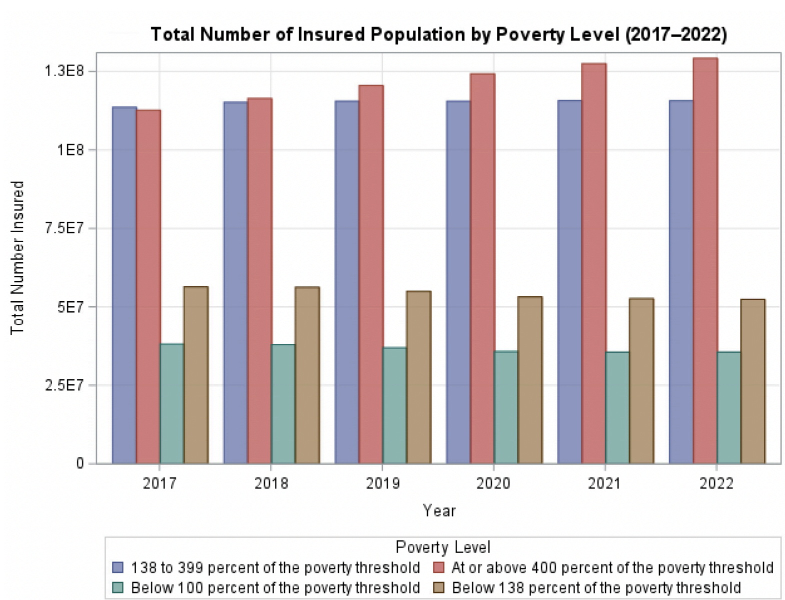


Figure 13: Total Number of Uninsured Population by Poverty Level (2017-2022)

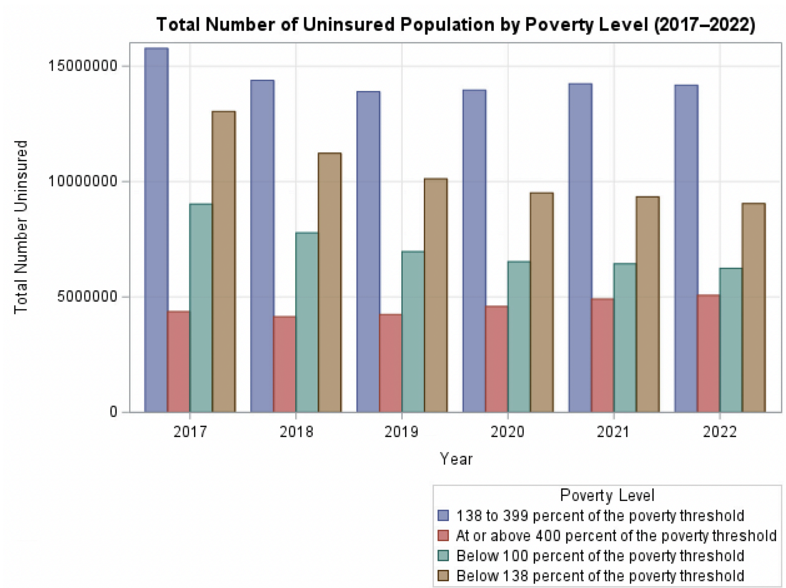
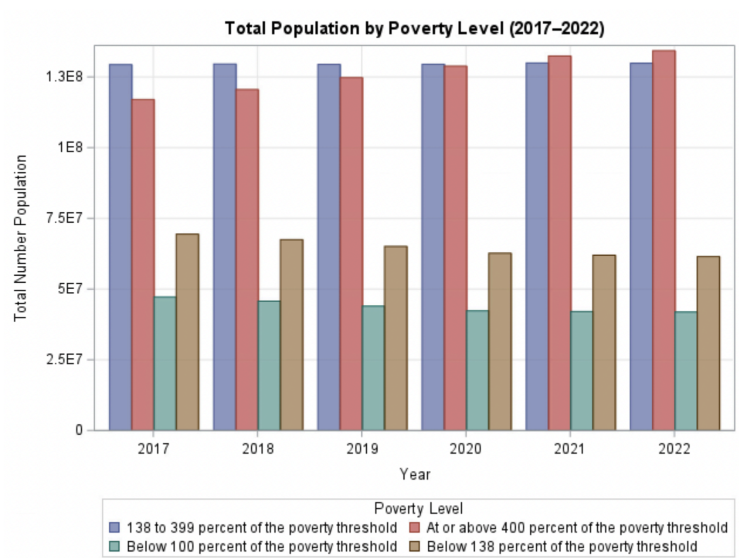


Figure 14: Total Population by Poverty Level (2017-2022)



Ratio of Income to Poverty Level:

Figures 12, 13, and 14 provide detailed insights into trends in health insurance coverage in the United States, categorized by poverty levels from 2017 to 2022. These figures focus on three important aspects: the total number of insured individuals, uninsured individuals, and the total population in each poverty category. The analysis highlights the overall trends, and comparisons among different poverty groups, and provides key conclusions about health insurance coverage over this six-year period.

The insured population shows a steady increase across all poverty categories from 2017 to 2022. Individuals "At or Above 400% of the Poverty Threshold" have the largest insured population throughout the period. The number of insured people in this group rises from approximately 112 million in 2017 to about 129 million in 2022. This steady increase suggests that people in higher-income categories have better access to health insurance due to factors such as employer-provided plans or their ability to afford private insurance. Similarly, the group "138% to 399% of the Poverty Threshold" also experiences significant growth in the insured population, from around 112 million in 2017 to 127 million in 2022. This indicates the success of public policies such as the Affordable Care Act (ACA), which supports middle-income individuals with subsidies and expanded coverage options.

On the other hand, the growth in the insured population for lower-income groups is slower. For individuals "Below 138% of the Poverty Threshold," the insured population increased modestly, from 54 million in 2017 to 60 million in 2022. Similarly, for those "Below 100% of the Poverty Threshold," the insured population grew from approximately 39.6 million to 43 million. This slow progress highlights the challenges faced by low-income populations, who are more dependent on government programs such as Medicaid. However, Medicaid coverage eligibility is not uniform across states, which might limit access for many individuals in these categories.

The uninsured population shows a gradual decline across all poverty levels. Individuals "At or Above 400% of the Poverty Threshold" have the lowest uninsured numbers, decreasing from 4.3 million in 2017 to about 3.9 million in 2022. This suggests that people with higher incomes face fewer barriers to accessing health insurance. For the group "138% to 399% of the Poverty Threshold," the number of uninsured individuals also shows a noticeable decline, from 15.6 million in 2017 to around 14.1 million in 2022. These trends reflect the positive impact of government policies, employer-sponsored plans, and subsidies in helping these individuals obtain insurance coverage.

However, uninsured rates among lower-income groups remain high, despite some improvements. For individuals "Below 138% of the Poverty Threshold," the uninsured population decreased from 12 million in 2017 to 9.1 million in 2022. Similarly, for those "Below 100% of the Poverty Threshold," the uninsured numbers reduced from 8.9 million in 2017 to 6.1 million in 2022. While there is progress, these groups still face significant barriers to obtaining insurance coverage. This may be due to limited Medicaid expansion in some states, as well as administrative challenges and affordability issues for the lowest-income populations.

The total population figure across poverty levels remains relatively stable during this period, which helps in understanding the overall trends. Higher-income groups ("At or Above 400% of the Poverty Threshold") consistently have higher insured rates and lower uninsured rates, showing a direct link between income level and access to health insurance. Meanwhile, lower-income groups ("Below 138%" and "Below 100% of the Poverty Threshold") show slower improvements, which reveals ongoing inequalities in access to healthcare coverage.

In conclusion, the figures show progress in increasing the overall number of insured individuals and reducing the uninsured population over time. However, the data also reveals significant disparities between poverty levels. Higher-income individuals enjoy much better access to health insurance, benefiting from employer-sponsored plans and their ability to afford private coverage. Meanwhile, lower-income populations continue to face challenges due to systemic barriers and limited policy support. To address these inequalities, efforts such as expanding Medicaid, simplifying the enrollment process, and providing greater subsidies for low-income individuals are necessary. This analysis underscores the importance of ongoing policy interventions to ensure equitable access to health insurance for all socioeconomic groups in the United States.

Interpretation of Results

Throughout our analysis, we delve into the distinct disparities among health insurance coverage gaps across age, disability status, education, work experience, income levels, and their various subcategories. From our statistical analysis and the models we created, numerous areas within each of our variables displayed consistent and relational coverage gaps. Most notably, older individuals, ages 65 and older, benefitted from high coverage rates due to Medicare, whereas middle age groups, those ages 19 to 64 faced much higher insurance coverage gaps, tied to employment status and eligibility for public programs. Additionally, individuals who had disabilities revealed a gradual improvement in coverage over time that can be attributed to Medicaid and Medicare expansion. However, there were still displays of disparities which can be attributed to differences in Medicaid policies across states. Moreover, education levels were discovered to be significantly associated with insurance coverage, as those with higher education levels maintained higher employer-sponsored plans, whereas, on the other hand, lower education levels were associated with lower levels of coverage due to reliance on public assistance and remained at an increased risk of being uninsured.

Within the entirety of our analysis, we came to the conclusion that significant drivers of health insurance coverage gaps among a variety of subcategories were employment and income. Full-time workers demonstrated the highest insurance rates from employer-sponsored plans, while unemployed and part-time individuals experienced a decline in coverage, displaying their vulnerability. Income disparities were particularly pronounced, as high-income earners enjoyed basically universal coverage whilst the opposite for low-income individuals. Policy measures such as Medicaid expansion and the Affordable Care Act, also can be attributed to trends among each variable with reduced uninsured rates for certain groups, however, significant gaps do remain. These results underscore the importance of addressing systemic barriers that impact low-income and underemployed populations.

Conclusion

Overall, while progress has been made in expanding insurance coverage for individuals across the US from 2017-2022, there still remain significant disparities that persist among various socioeconomic factors (Keisler-Starkey, 2022). High-income individuals and full-time workers have consistently benefited from targeted policy interventions and employer-sponsored plans, for better coverage rates, as highlighted in Figures 10 and 8. However, middle-aged adults, individuals with disabilities, lower-income individuals, and part-time workers remain heavily reliant on public programs, with many still uninsured, as shown in Figures 9, 11, and 13 (Carman et al., 2015; Cha & Cohen, 2022). These gaps highlight the divergent access to affordable health insurance and the limitations in the current systems, particularly for low-income populations and those facing unstable employment (Carman et al., 2015).

In order to achieve equitable health insurance coverage, expanding and updating health insurance policy measures are imperative. For example, expanding Medicaid eligibility, enhancing ACA subsidies for those in low, middle-income groups, and also implementing reforms to decrease private insurance costs that would significantly lower uninsured totals (Buettgens & Ramchandani, 2023). Furthermore, addressing issues with barriers faced by unemployed and part-time workers could dramatically aid in bridging the health insurance gap, as reflected in figures 8 and 9. These steps, combined with reforms to decrease private insurance costs, could help mitigate existing inequities. As this report emphasizes, comprehensive and inclusive strategies are vital to ensuring that individuals, regardless of age, employment status, or income, have affordable access to health insurance coverage (Vardell, 2019; Cha & Cohen, 2022).

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Appendix A: Data Sources and Compilation Process

Within our analysis, we utilized comprehensive datasets from the U.S. Census Bureau, which focused on insurance health coverage from 2017 to 2022. For our primary data points, we included all 50 states, including D.C. to ensure national coverage, and socio-demographic variables of age, disability status, educational attainment, work experience, and poverty levels as a total. Additionally, variables were grouped into subcategories as totals for both insured and uninsured individuals: age groups for under 19, 19-64, and 65 and older, income levels as $100,000 and over, $75,000 to $99,999, $50,000 to $74,000, $25,000 to $49,999, and under $25,000, poverty thresholds for below 100%, 13-399%, and 400% and over, work status, unemployed, part-time, and full time, and lastly disability status for not disabled and disabled.

Moreover, we accounted for incomplete or missing data to ensure consistency in year-to-year comparisons. Once we compiled our data for each variable and their subcategories totals as well as percentage breakdowns over time, our data processing was performed using SAS for analysis, cleaning, and visualizations. Our descriptive statistics provided the comprehensive foundation for identifying the key predictors of observations, mean, lower, and upper 95% confidence interval for mean and standard deviations for insurance coverage gaps. We illustrated these insights through detailed figures that showcased each of our groups and their subcategories within insured and uninsured populations. This approach allowed us to create an in-depth cross-comparison, further highlighting significant trends over time and disparities within and across each socio-demographic group.

Appendix B: Heat Map Analysis

In this appendix, we have presented two sets of additional heatmaps to provide a comprehensive visualization of the distribution of the total population in two categories i.e.

insured represented by Appendix (B1), and uninsured represented by Appendix (B2) across the US counties from 2017 to 2022. These maps were created to better understand the trends and geographical distribution of individuals with and without health insurance coverage over time.

Appendix (B1) illustrates the yearly insured populations across U.S. counties from 2017 to 2022. Over time, there is a noticeable shift towards darker orange shades, which represent higher numbers of insured individuals. This indicates a steady increase in health insurance coverage across most counties, reflecting improvements in accessibility or adoption of health insurance during this period.

Appendix (B1): Distribution of Insured People in the USA by County

Appendix (B2) depicts the yearly uninsured populations by county for the same period. Here, the trend moves in the opposite direction, with lighter shades, representing fewer uninsured individuals, becoming more prominent over the years. This suggests a significant reduction in the uninsured population across the U.S., particularly in counties with historically higher uninsured rates.

Appendix (B2): Distribution of Uninsured People in the USA by County

By examining these visualizations, we can observe clear shifts in coverage patterns and analyze how insurance accessibility has evolved geographically across the U.S. from 2017 to 2022. These insights help contextualize broader discussions about the effectiveness of healthcare policies and coverage initiatives, taking into account significant variables and their subgroups among the displayed heatmaps, during this period which we have discussed in the paper.